

Immunology

Ingyu Bahng

August 2026

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1 Antibodies

1.1 Immunoglobulin Structure & Isotypes

Core Concept 1.1: Immunoglobulin Superfamily

The **immunoglobulin superfamily** refers to a group of molecules containing at least one **immunoglobulin (Ig) domain**. Regardless of the specific protein it belongs to, almost all domains share a highly conserved 3D structure consisting of 70 to 110 amino acids folded into a compact lump of **β pleated sheets** that are pinned together by a single **intradomain disulfide bond** between **two cysteine residues**

It's natural confirmation results in the inner-facing side chains of the β sheets being primarily **hydrophobic**, driving them to pack itghly against each other, further stabilizing what is often referred to as the "sandwich" structure.

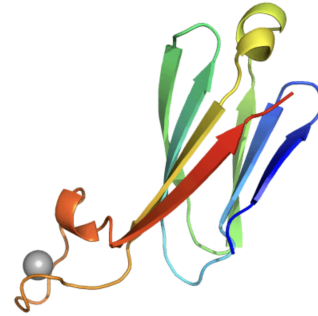


Figure 1: Three-dimensional structure of an immunoglobulin domain

Core Concept 1.2: Immunoglobulin Receptors & Antibodies

An **antibody**—also known simply as an **immunoglobulin** or **Ig**—is a highly specialized, Y-shaped protein composed of Ig domains that is produced by the immune system to recognize and neutralize foreign threats. An **Ig receptor** (or *B cell receptor*) is an antibody bearing a membrane-spanning and cytosolic domain at its *C-terminal* end. The general antibody structure can be divided into two **light (L)** and two **heavy (H)** chains.

- (a) **Light Chains:** The two light chains— 25, 000 MW; 50, 000 MW total for both—are identical to one another and each consist of 2 Ig domains: one **variable** (V_L in Figure 3) and one **constant** (C_L in Figure 3).

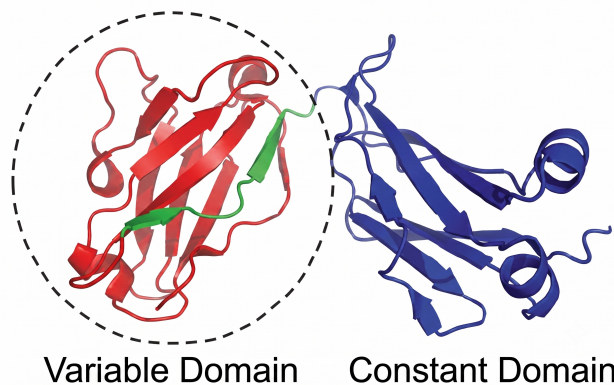


Figure 2: Three-dimensional structure of an antibody light chain; variable Ig domain (left), joining segment (green), constant domain (blue)

- **Constant Region (C_L):** The carboxy-terminal (COOH) portion of this region **links with the heavy chain**, contributing to the hinge structure that forms the base of the characteristic Y-shape. Two distinct isotypic variants of the light chain constant region exist: the **kappa (κ)** and **lambda (λ)** chains, the latter of which can be further divided into five structural subclasses.

- **Variable Region (V_L):** This region contains three distinct loops extending from the terminus that together form the **hypervariable region**, constructed from three non-contiguous amino acid sequences and comprising approximately 15–20% of the domain.

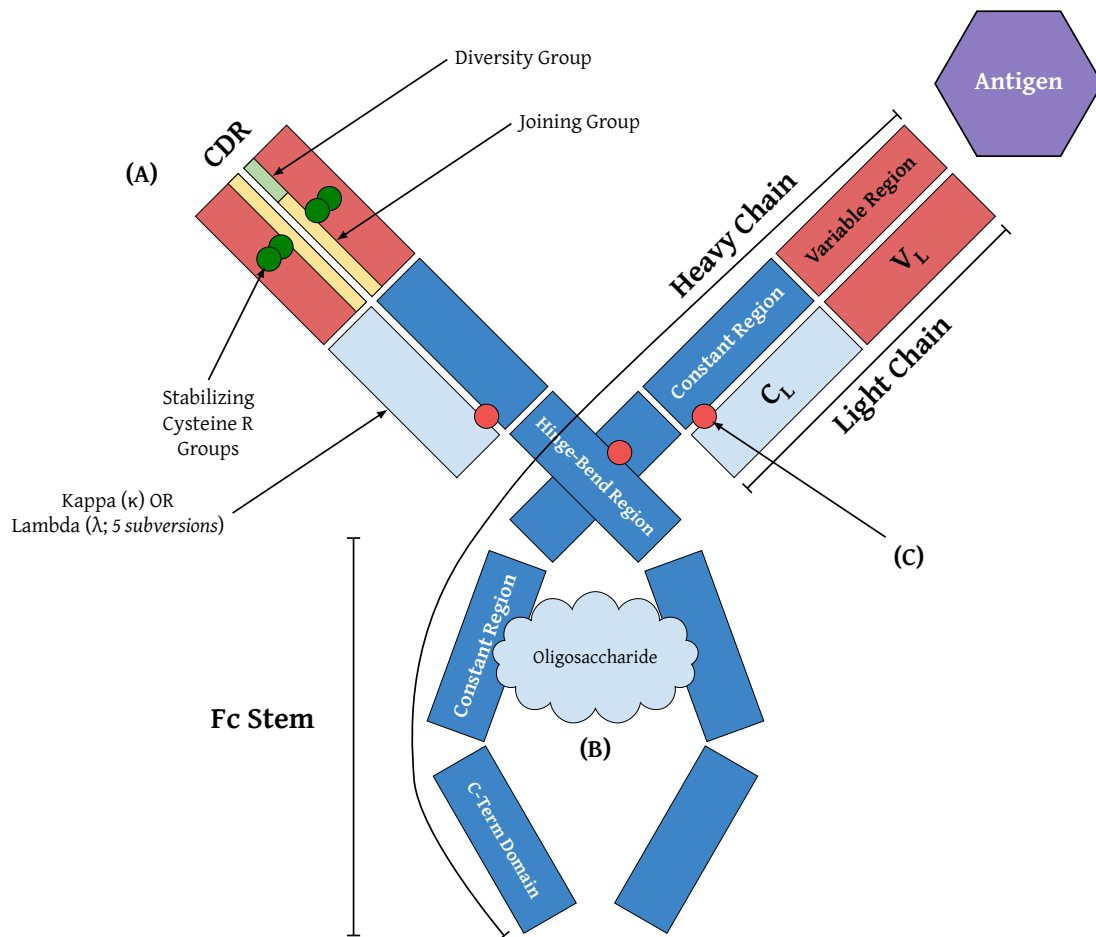


Figure 3: Antibody structure diagram; (A) **three loops** protrude from the ends of the variable domains and serve as the site of connection for a particular **antigen epitope**—each distinct CDR site is referred to as a distinct **idiotype**; (B) an **oligosaccharide attaches to the second Ig domain**, determining how the antibody interacts with the rest of the immune system; (C) both light-heavy and heavy-heavy linkages involve **covalent disulfide bonds** and strong **non-covalent forces**

- (b) **Heavy Domain:** The two heavy chains— 50,000 MW; 100,000 MW for both—are also identical to one another and each consist of 4 or 5 Ig domains: one variable domain and the remaining constant domains, which serve the same general function across distinct antibodies.

- **Variable Region (V_H):** Like the V_L domain, three terminal loops form the **hypervariable region** of the heavy chain. When spatially paired with the three corresponding loops of the light chain V_L , these loops collectively form the **complementarity-determining regions (CDRs)**, which are responsible for unique, hyperspecific antigen binding.
- **First Constant Domain**
- **Hinge-Bend Region:** This region is either a flexible **hinge** or a rigid **bend**, depending on the specific **Ig isotype** (Core Concept 1.3). In IgM and IgE (possessing μ and ϵ heavy chains, respectively), this region consists of additional, highly rigid immunoglobulin domains. In IgG, IgA, and IgD (possessing γ , α , and δ heavy chains), it instead consists of a proline-rich amino acid sequence stabilized by disulfide bonds,

which provides substantial structural flexibility.

- **Next Constant Domain:** This domain serves as the primary site for post-translational **oligosaccharide attachment**, which spatially opens the domains to the aqueous environment, which is essential for facilitating subsequent interactions.
- **Carboxy-Terminal (C-term) Domain:** This terminal domain dictates the antibody's final localization: whether it is retained as a **membrane-bound receptor** (*on naive or memory B cells*) or **secreted extracellularly** (*by plasma cells*). The secreted variant terminates in a short, exclusively hydrophilic amino acid sequence, whereas the membrane-bound variant terminates in a specialized sequence containing a hydrophobic transmembrane span followed by a short, cytoplasmic hydrophilic tail.

Core Concept 1.3: Immunoglobulin Isotypes (H Chain Differences)

There are five classes of antibodies—termed IgM, IgD, IgG, IgA, and IgE—which differ in function as well as in the exact amino acid sequence and conformation of the Fc stem. In all cases, the basic unit consists of two heavy chains and two light chains, and the light chain may be either κ or λ , as discussed in Core Concept 1.2. The classification of antibodies into these five classes depends on the structural elements of and functions dependent on the heavy chain as outlined below.

- **Flexible hinge/Rigid bend:** The number of constant (C) Ig domains present, which determines whether the heavy chain possesses a flexible hinge region or a more rigid bend.
- **Oligosaccharide:** A small carbohydrate group is added to the second domain from the C-terminus, with the specific carbohydrate varying according to antibody type.
- **J Chain:** The presence of a J chain allows certain antibodies to form compound structures capable of crossing epithelial layers.
- **Subclasses:** Distinct subcategories exist within a given antibody class (*e.g., IgA1, IgA2, IgA3, IgA4*).

IgX	Hinge/ J chain Bend		Sub-classes	Timing	Membrane Spanning
M	Rigid	✓	—	First class produced in maturing B cells	Naïve & memory
D	Hinge	×	—	Produced as Ig receptor on mature, naïve B cells	Naïve cells (<i>rarely soluble/memory</i>)
G	Hinge	×	4	After class switching in activated B cells	Memory cells
A	Hinge	✓	2	After class switching in activated B cells	Memory cells
E	Bend	×	1	After class switching in activated B cells	Memory cells

Table 1: Structural and functional properties distinguishing the five immunoglobulin classes.

The following is a more in depth look into the functional roles and characteristics of each Ig subclass.

(a) **IgM** (*Memory & Naive B Cells*)

IgM is structurally defined by a **rigid bend** region comprising four constant domains and serves as a critical

first-line defense, representing the first antibody isotype secreted into plasma during a **primary immune response**.

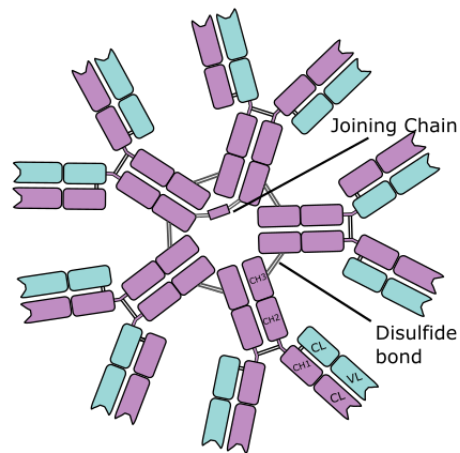


Figure 4: IgM Pentamer

When expressed as a membrane-bound B cell receptor, it exists strictly as a monomer; however, its secreted soluble form assembles into a large **pentameric structure** possessing ten distinct antigen-binding sites. This pentamer is stabilized by a central **J chain**, which both aids in **mucosal secretion** and, through its multimeric geometry, renders IgM highly efficient at **cross-linking antigens** and initiating the **complement cascade**.

As with IgG, this potent complement activation drives rapid **opsonization** of targets and promotes their subsequent phagocytic destruction. Despite its larger size, IgM can also undergo **transcytosis** across mucosal barriers.

(b) **IgD** (*Naive B Cells*)

IgD possesses a **flexible hinge** structure and assists in the **initial recognition of antigen by naive B cells**, predominantly operating as a **membrane-bound B cell receptor** rather than as a secreted antibody; accordingly, it is detected in plasma only at trace levels, accounting for roughly 0.2% of total serum immunoglobulins.

(c) **IgG** (*Memory B Cells*)

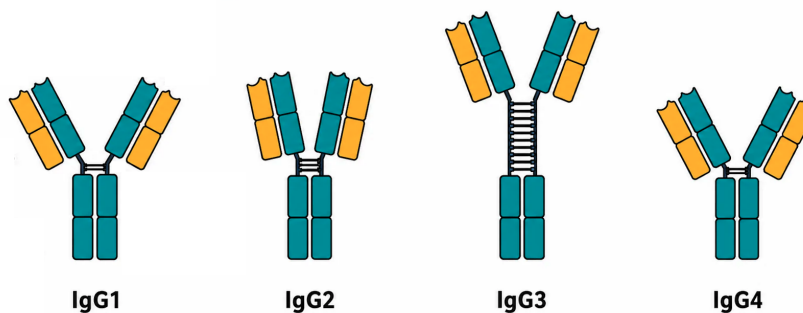


Figure 5: IgG subclasses (*IgG1, IgG2, IgG3, and IgG4*)

IgG is characterized by a **flexible hinge** structure comprising three constant domains and represents the **predominant secreted antibody class**, responsible for widespread systemic defense against both bacterial and viral pathogens. It is highly effective at activating the **complement system**, a process that leads to pathogen **opsonization** and consequently enhances phagocytic clearance. IgG further diversifies into four subclasses, numbered 1 through 4, which vary in their biological specificity and effector function, as summarized below.

Subclass	Complement Activation	Fc Receptor Binding
<i>IgG1</i>	Moderately activates	Binds tightly
<i>IgG2</i>	Weakly activates	Binds weakly
<i>IgG3</i>	Strongly activates	Binds tightly (high disulfide content)
<i>IgG4</i>	Does not activate	Binds weakly

Table 2: Comparative complement activation and Fc receptor binding across IgG subclasses.

(d) **IgA** (*Memory B Cells*)

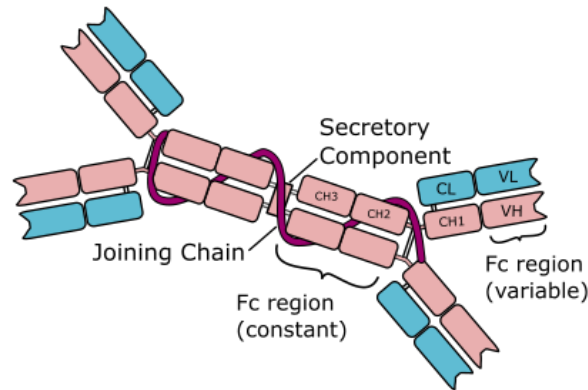


Figure 6: IgA dimer

IgA possesses a **flexible hinge** structure and exists in two subclasses, IgA1 and IgA2. Its primary role involves **immune surveillance** and **defense at mucosal and epithelial barriers**, and it is secreted in substantial quantities—up to 15 grams per day—into mucus, tears, saliva, and breast milk. While IgA circulates in the serum **largely as a monomer**, it is **assembled and secreted across epithelia primarily as a dimer**, covalently linked by a **J chain** that facilitates its secretion into **mucus layers** and localizes the antibody to various **epithelial linings**; the active transport mechanism responsible for moving these antibodies across epithelial cell layers is termed **transcytosis**.

(e) **IgE** (*Memory B Cells*)

IgE possesses a **rigid bend** region owing to an additional constant domain, and as a monomer it superficially resembles IgM, though it differs somewhat in amino acid sequence. Physiologically, it is adapted for immunological defense against **large parasitic helminths, or worms**. IgE is also deeply implicated in allergic pathogenesis, as its constant regions bind with exceptionally **high affinity to specialized Fc receptors** on the surface of mast cells and basophils, triggering the **characteristic allergic response**.

1.2 Immunoglobulin Expression

Core Concept 1.4: Immunoglobulin DNA to Expression

<FIGURE>

Core Concept 1.5: Immunoglobulins in Action

Antigen recognition occurs at the six loops located at the tips of the antibody’s Y-shaped arms, mediated by the same weak, non-covalent interactions responsible for enzyme-substrate binding. As in enzyme-substrate interactions, this binding process can involve induced fit, in which both the antibody and the antigen undergo conformational distortion to achieve optimal complementarity.

Antigens are usually proteins, and several structural features govern how B-cell epitopes are recognized. Epitopes are typically located at the surface of a protein, often corresponding to regions that protrude outward and are thus most accessible to antibody binding. The protein’s tertiary structure, or three-dimensional shape, is critical to this recognition; denatured proteins consequently fail to bind, since the epitope’s native conformation has been lost. Quaternary structure can also play a role, as an antibody may recognize an epitope formed at the junction between two distinct associated peptides. Because of this dependence on three-dimensional conformation, B-cell epitopes may be formed either by contiguous (sequential) stretches of amino acids or by residues that are non-sequential in the primary structure but are brought into spatial proximity through protein folding.

1.3 Immunoglobulin Genetics

Because somatic cells are diploid, they possess two alleles for every relevant chromosome. Consequently, each genetic rearrangement event has the potential to occur on either allele within the genome. The genetic loci encoding immunoglobulins consist largely of tandemly repeated, homologous segments that function as interchangeable coding blocks during somatic recombination.

Core Concept 1.6: L Chain Gene Sequences

(a) λ L Chain

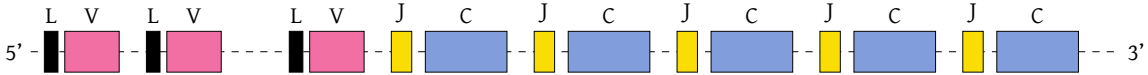


Figure 7: caption

Localized to Chromosome 22. V Segments: Approximately 30 to 40 distinct coding blocks. J Segments: 4 distinct coding blocks. C Segments: 5 blocks total, containing 2 functional exons (Cλ1, Cλ2) and 3 pseudogenes (Cλ3, Cλ4, Cλ5).

(b) κ L Chain

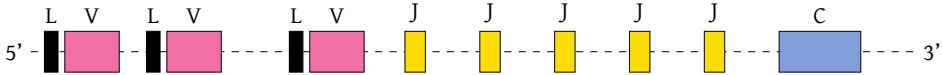


Figure 8: caption

Localized to Chromosome 2. V Segments: Approximately 35 to 40 distinct coding blocks. J Segments: 5

distinct coding blocks. C Segments: 1 functional coding block (C κ).
<FIGURE details>
Further molecular details concerning these sequence loci will be elaborated upon below.

Core Concept 1.7: H Chain Gene Sequences

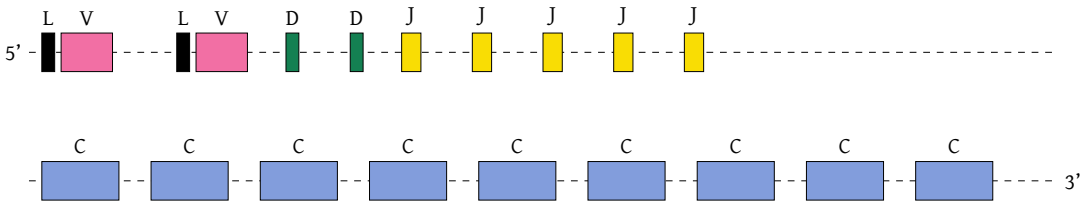


Figure 9: caption

Localized to Chromosome 14.

V Segments: Approximately 40 functional segments.

D Segments: Approximately 23 functional segments.

J Segments: Approximately 6 functional segments.

C Segments: 8 distinct domains that correspond directly to the various antibody isotypes (noting that the closely linked C μ and C δ segments are functionally transcribed together prior to alternative splicing).

The specific downstream genomic arrangement of these constant regions follows the order: C μ /C δ , C γ 3, C γ 1, C α 1, C γ 2, C γ 4, C ϵ , and C α 2.

<FIGURE details>

Examining the heavy chain constant regions at a higher resolution reveals an organized substructure of coding sequences:

Each region contains 3 to 4 independent exons that correspond to the individual globular domains comprising the structural heavy chain of that isotype.

Situated downstream of the primary structural exons are two highly conserved sequences coding for the hydrophobic membrane-spanning extensions.

Importantly, separate polyadenylation (poly-A) signal sites exist after both the secreted-sequence exons and the membrane-spanning exons, dictating through alternative RNA processing whether the final synthesized antibody will be secreted or membrane-anchored.

1.3.1 L/H Chain Gene Expression

Core Concept 1.8: λ L Chain Gene Expression

(1) REARRANGEMENT [Rearranged Gene] — Nucleus

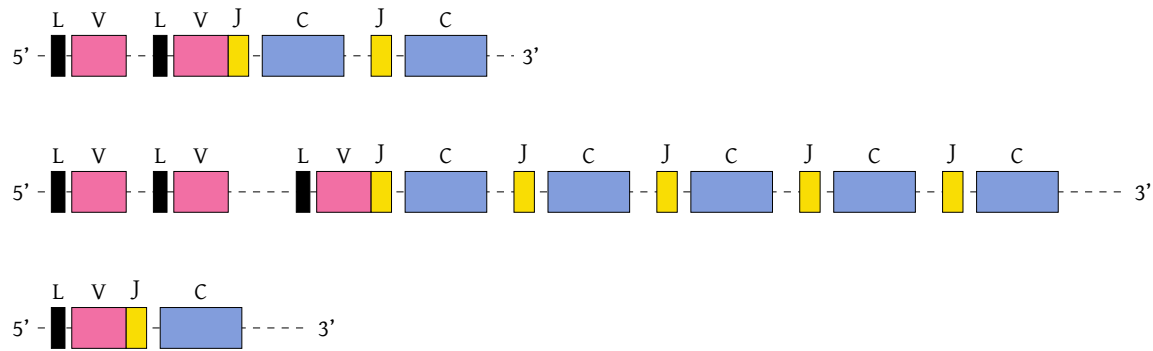


Figure 10: caption

A single variable (V) segment undergoes somatic recombination with a single joining (J) segment. This DNA rearrangement physically excises any intervening genetic sequences and functionally activates only the promoter immediately upstream of the successfully joined V segment.

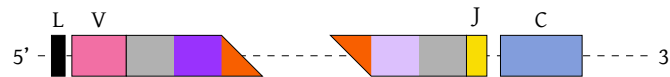


Figure 11: caption

This process relies on the Recombination Signal Sequence (RSS) mechanism detailed in a subsequent section.

(2) **TRANSCRIPTION** [Primary Transcript] — *Nucleus*

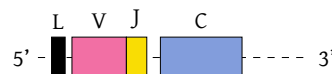


Figure 12: caption

RNA polymerase generates a primary transcript covering the rearranged VJ sequence, continuing through the downstream constant (C) region until it terminates at the designated transcription stop signal.

(3) **PROCESSING** [Processed Message] — *Nucleus*



Figure 13: caption

Spliceosomes process the primary RNA, removing non-coding intronic sequences located between the rearranged VJ unit and the C domain.

Post-transcriptional modification includes the addition of a poly-A tail at the 3' end to ensure messenger RNA stability.

(4) **TRANSLATION** [Nascent Peptide] — *Into the RER*



Figure 14: caption

Ribosomal complexes bind the mature mRNA in the cytosol. Upon synthesizing the initial targeting sequence, the ribosome docks to the Rough Endoplasmic Reticulum (RER), allowing the nascent polypeptide chain to be translocated directly into the ER lumen.

This docking is mediated by a hydrophobic leader (L) sequence at the N-terminus of the developing protein, which interacts with signal recognition particles.

(5) **L REGION SNIPPING** [Mature Chain] — ER Lumen



Figure 15: caption

Signal peptidases within the ER lumen cleave the N-terminal leader sequence, producing a mature light chain polypeptide consisting solely of the variable (VJ) domain and the constant (C) domain.

Core Concept 1.9: κ L Chain Gene Expression

(1) **REARRANGEMENT** [Rearranged Gene] — Nucleus

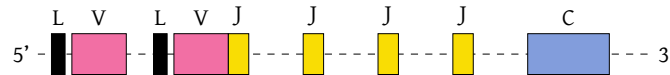


Figure 16: caption

Similar to the lambda chain, a single V segment undergoes recombination with a single J segment. Any intervening J sequences are permanently excised from the DNA, and the upstream promoter corresponding exclusively to the chosen V segment becomes functionally active.

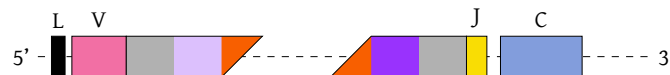


Figure 17: caption

This DNA rearrangement is mediated by the RSS molecular mechanism.

(2) **TRANSCRIPTION** [Primary Transcript] — Nucleus



Figure 18: caption

Transcription initiates at the active promoter, generating a primary RNA transcript that spans the chosen

V segment, the selected J segment, any remaining downstream unselected J segments, and the single C κ constant region.

(3) **PROCESSING** [Processed Message] — *Nucleus*



Figure 19: caption

RNA splicing enzymes excise all intronic sequences and the unselected downstream J segments, yielding a contiguous coding message.

A poly-A tail is appended to stabilize the transcript for cytosolic export.

(4) **TRANSLATION** [Nascent Peptide] — *Into the RER*



Figure 20: caption

This process proceeds identically to lambda light chain translation, involving RER docking.

(5) **L REGION SNIPPING** [Mature Chain] — *ER Lumen*



Figure 21: caption

Proteolytic removal of the leader sequence within the ER yields the final mature protein, comprising the variable and constant regions.

Core Concept 1.10: H Chain Gene Expression

(1) **REARRANGEMENT** [Rearranged Gene] — *Nucleus*

The initial recombination event joins one Diversity (D) segment to one J segment. All intervening D and J segments are excised from the genome, while those situated downstream remain intact.

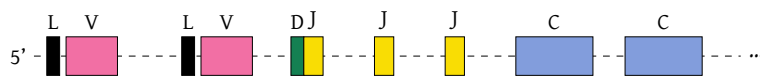


Figure 22: caption

Following the D-J joining, a second recombination event splices a V segment to the newly formed D-J complex. This physically deletes all unselected V segments located downstream of the chosen V segment, and any D segments located upstream of the chosen D segment.

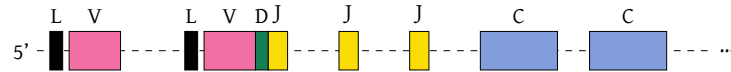


Figure 23: caption

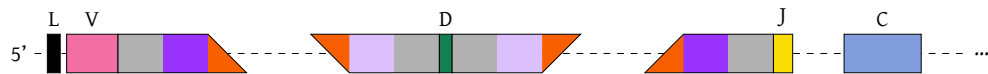


Figure 24: caption

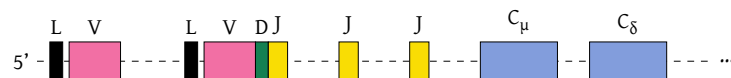
(2) **TRANSCRIPTION** [Primary Transcript] — *Nucleus*

Figure 25: caption

The primary transcript synthesized spans the fully rearranged V-D-J sequence, encompassing all downstream introns and remaining J segments. Importantly, transcription continues unabated through both the C_μ and C_δ constant region coding domains, terminating only after the poly-A site situated downstream of C_δ .

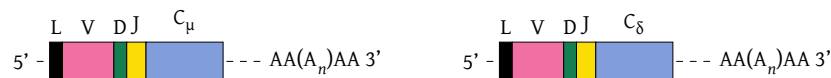
(3) **PROCESSING** [Processed Message] — *Nucleus*

Figure 26: caption

The elongated primary transcript is subject to alternative RNA splicing, a critical regulatory step determining whether the mature mRNA will ultimately incorporate the exons for the μ heavy chain (forming IgM) or the δ heavy chain (forming IgD), but not both simultaneously on the same transcript.

Concurrently, all intronic sequences and superfluous unselected J segments are spliced out.

(4) **TRANSLATION** [Nascent Peptide] — *Into the RER*

Figure 27: caption

Follows the same ER-docking and luminal translocation process described for light chains.

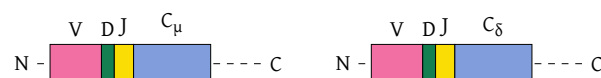
(5) **L REGION SNIPPING** [Mature Chain] — *ER Lumen*

Figure 28: caption

Proteolytic removal of the leader sequence yields the mature heavy chain polypeptide, featuring the highly variable V-D-J domain and the selected constant isotype domain.

Core Concept 1.11: RSS Joining Mechanism

(a) Juxtaposition

- *Locating & Cleaving*

Target loci destined for rearrangement physically relocate to transcriptionally active zones within the nucleus. The chromatin structure relaxes, displacing nucleosomes to grant enzymatic access to the naked DNA.

The RAG-1/RAG-2 recombinase complex aligns specific Recombination Signal Sequences (RSS), strictly adhering to the "12/23 rule," where a sequence featuring a one-turn spacer must pair with a sequence featuring a two-turn spacer. Upon alignment, an endonucleolytic nick is introduced precisely at the 5' border of the conserved heptamer sequence.

<FIGURE>

- *Hairpin Formation*

The newly liberated 3' hydroxyl group executes a nucleophilic attack on the complementary phosphodiester bond of the intact strand. This transesterification reaction creates a covalently sealed hairpin loop at the coding ends of the participating V and J segments (in light chains), or the V, D, and J segments (in heavy chains).

Conversely, the excised signal ends—containing the heptamer, spacer, and nonamer sequences—are left as a blunt, flush cut featuring a free 5' phosphate group.

<FIGURE>

(b) Ligation

- *Initial Cleavage & Ligation*

<FIGURE>

The Artemis nuclease complex resolves the coding hairpins through an asymmetric, random single-stranded cleavage. This imprecise nicking generates varying lengths of unmatched single-stranded DNA overhangs at the joining interface.

- *P-Nucleotide Addition*

<FIGURE>

Cellular repair polymerases fill in these asymmetrical overhangs, synthesizing short palindromic (P) nucleotide sequences that conform perfectly to the templates created by the offset hairpin resolution.

- *N-Nucleotide Addition (Heavy Chains Only)*

<FIGURE>

In developing heavy chains, the enzyme Terminal Deoxynucleotidyl Transferase (TdT) can independently append up to 15 non-templated (N) nucleotides randomly into the junctional gap prior to final ligation, profoundly expanding the sequence diversity of the resulting antigen receptor.

(c) Result

The culmination of this process yields a fully repaired and ligated coding joint, forming the functional antigen-binding domain. Meanwhile, the excised signal sequence fragment—typically forming a closed episomal circle consisting of the joined heptamer-spacer-nonamer elements—is generally degraded, unless specific structural topologies dictated an inversional joining event.

Subsequent to baseline V(D)J recombination, mature B cells undergo Somatic Hypermutation. During active immune responses, point mutations are intentionally introduced into the hypervariable gene loci, providing the substrate for affinity maturation.

It is essential to delineate that somatic hypermutation occurs exclusively in peripheral secondary lymphoid tissues following antigen stimulation, well after the initial B cell maturation phase in the bone marrow has concluded.

1.4 Other

Core Concept 1.12: *Edelman & Porter*

In foundational experiments conducted during the 1960s, researchers utilizing serum protein electrophoresis fractionated human plasma into five distinct molecular cohorts. One specific migratory band, designated the "*gamma*" globulin fraction, was identified as housing the vast majority of the plasma's protective immunological activity. Subsequent purification of this fraction **led to the biochemical isolation and characterization of antibody molecules.**

(a) **Gerald Edelman** (*Mercaptoethanol*)

Edelman employed **mercaptoethanol** to **reduce the disulfide bonds** that keep the light and heavy chains interconnected with one another, thereby enabling each chain to be isolated and studied individually.

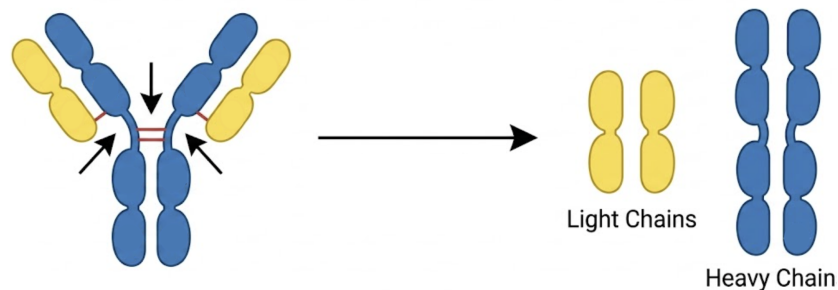


Figure 29: Antibody cleavage via mercaptoethanol; *cleavage occurs at disulfide bonds, reducing the antibody to individual light and heavy chains*

(b) **Rodney Porter** (*Pepsin & Papain*)

Porter demonstrated that **pepsin** cleaves preferentially at the **bottom of the hinge** between the arms and the stem, thereby disconnecting the two individual heavy chain Fc sections from the top half of the molecule and yielding a single **Fab2 fragment**.

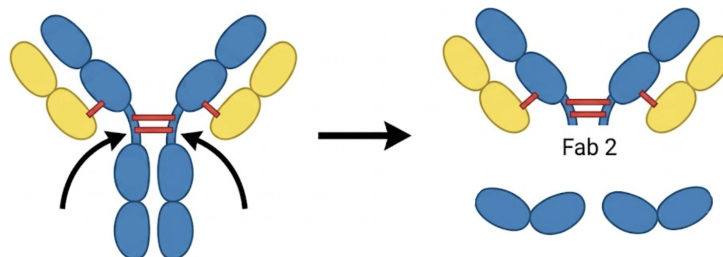


Figure 30: Antibody cleavage via pepsin; *cleavage occurs at hinge, separating two heavy chain fragments and one Fab2 molecule*

By contrast, he found that **papain** cleaves preferentially at the **top of the hinge** between the arms and the stem, disconnecting the Fc sections—which remain joined to one another by disulfide bonds—from two separate Fab fragments.

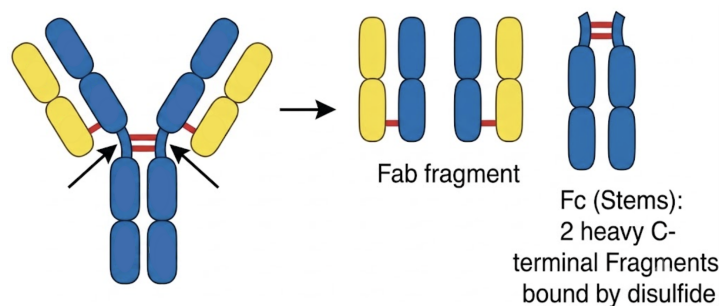


Figure 31: Antibody cleavage via papain; cleavage occurs at upper hinge, resulting in connected lower heavy chain fragments and two Fab fragments

Notably, structurally **divalent Fab2 fragments** (from *pepsin* treatment) remain capable of precipitating or cross-linking antigens to form extended immune complexes, whereas **monomeric Fab fragments** (from *papain* treatment) lack this cross-linking ability.

Core Concept 1.13: Haptens (Small Antigens)

Haptens are **low-molecular-weight** chemical entities that can be recognized by antibodies but are **inherently non-immunogenic**; their minute size prevents them from independently eliciting an adaptive immune response.

To overcome this limitation for experimental or clinical antibody production, haptens must be covalently conjugated to a **high-molecular-weight** carrier protein, such as **bovine serum albumin (BSA)**, to provide the requisite steric bulk and T cell epitopes needed to induce active immunity.

<FIGURE>

Core Concept 1.14: Monoclonal Antibodies (mAb)

The ability to engineer continuous, uniform production of antibodies featuring customized, highly specific CDR loops targeting defined antigen epitopes holds profound clinical and pharmacological value, especially concerning targeted oncological therapeutics.

Definition: Monoclonal antibodies represent a homogenous population of immunoglobulins synthesized by a single clonal lineage of B cells, resulting in absolute uniformity of paratope structure and antigen specificity.

This production is achieved through the generation of hybridomas—engineered, immortalized cell lines formed by fusing specific primary B cells with myeloma cells, thereby ensuring an indefinite, robust yield of the desired monoclonal antibody construct. The general synthesis methodology is detailed below.

<FIGURE>

2 B Cells

2.1 Naive B Cell Development

Core Concept 2.1: Naive B Cell Development

(1) Co-Receptor Development

<FIGURE>

The exposure of Common Lymphoid Progenitors (CLPs) to B-Cell Specific Activator Protein (BSAP) induces the initial surface expression of essential BCR signaling co-receptors, specifically $Ig\alpha$ and $Ig\beta$, firmly committing the precursor to the B lymphocyte developmental trajectory.

At this stage of commitment, the developing cell is classified as a progenitor B (pro-B) cell.

BSAP expression is maintained constantly across the entire B cell lineage continuum, ceasing only upon terminal differentiation into antibody-secreting plasma cells.

(2) Heavy Chain

<FIGURE>

The pro-B cell initiates V(D)J rearrangement at the heavy chain locus. The cell is afforded a maximum of two opportunities (one per allele) to generate a productive rearrangement before default apoptosis occurs.

Upon translating a viable heavy chain polypeptide, the nascent heavy chain associates with a pre-existing surrogate light chain complex, advancing the cell to the precursor B (pre-B) cell stage.

The functional assembly of this surrogate receptor complex delivers a crucial intracellular signal that enforces allelic exclusion at the heavy chain locus, maintains structural stability, and signals the nucleus to begin rearranging the endogenous light chain genes.

(3) Light Chain

<FIGURE>

The pre-B cell subsequently attempts V-J rearrangement of the light chain loci, possessing up to four total opportunities (spanning both the κ and λ alleles) before risking apoptosis.

A productive rearrangement yields a functional light chain that pairs with the heavy chain, resulting in the surface expression of a complete IgM BCR featuring a fully defined complementarity-determining region (CDR).

The cell, now bearing its unique surface receptor, transitions into the immature B cell designation.

(4) Negative Selection

Within the bone marrow microenvironment, immature B cells undergo negative selection to rigorously test their newly formed BCRs for potentially harmful autoreactivity against local self-antigens.

Approximately 10% of the initial developing B cell pool successfully navigates this strict tolerance checkpoint.

Autoreactive cells face elimination; binding of widespread multivalent self-antigens heavily crosslinks the surface IgM, triggering rapid apoptotic cell death pathways.

However, under specific conditions, an autoreactive B cell may avoid immediate destruction by reactivating Recombination Activating Gene (RAG) transcription, initiating a process termed receptor editing to generate a new, non-autoreactive light chain specificity.

Cells that successfully bypass negative selection exhibit functional central tolerance, lacking recognition for endogenous self-antigens.

(5) IgD Formation

<FIGURE>

Surviving cells employ alternative mRNA splicing to concurrently transcribe and translate the IgD isotype alongside their existing IgM receptors.

The dual expression of surface IgM and IgD signifies a fully mature, immunocompetent, yet naive B cell, which is now authorized to exit the bone marrow and enter systemic circulation.

(6) During Circulation

The mature B cells migrate through the circulatory and lymphatic systems to seed peripheral secondary lymphoid organs, including the lymph nodes, spleen, and various mucosal tissues.

Upon encountering specific antigen and receiving activating signals, the resulting plasma cells drastically upregulate transcription of the μ heavy chain, altering RNA processing to exclude the transmembrane coding domains, leading to massive secretion of the soluble, multimeric IgM pentamer.

Soluble secretion of the δ isotype (IgD) is physiologically negligible, as the cell rarely produces δ transcripts lacking the membrane anchor sequences.

2.2 B Cell Activation and Proliferation

2.2.1 Thymus Independent and Dependent B Cell Activation

Core Concept 2.2: Thymus Independent B Cell Activation

Certain distinct classes of antigens, termed Thymus-Independent (TI) antigens, possess the unique capability to induce B cell activation and proliferation in the complete absence of concurrent T helper (TH) cell stimulation. These structures frequently co-engage innate Toll-Like Receptors (TLRs).

TI-mediated activation is characterized by two fundamental limitations: it generally fails to induce functional immunoglobulin class switching (restricting output predominantly to IgM), and it does not yield a sustained population of memory B cells; robust induction of either process strictly requires classical TH cell involvement.

(a) Type 1

<FIGURE>

TI-1 antigens typically consist of potent mitogenic molecules, most notably lipopolysaccharide (LPS), derived from the outer leaflet of Gram-negative bacterial cell walls.

At significant concentrations, these molecules bind the BCR while simultaneously acting as potent ligands for adjacent TLR4 receptors, delivering the necessary, self-contained secondary permissive signal for cellular activation.

(b) Type 2

<FIGURE>

TI-2 antigens are characterized by highly repetitive, polymeric structural motifs (such as extensive bacterial capsular polysaccharides or flagellin polymers) that induce massive, sustained cross-linking and clustering of surface BCRs.

While lacking direct TH cell contact, successful TI-2 activation often relies upon a generalized, supportive microenvironment heavily supplemented by inflammatory cytokines (e.g., IL-2, IL-3, and IFN- γ) secreted by nearby innate immune populations and specialized APCs.

Core Concept 2.3: Thymus Dependent B Cell Activation

Under canonical circumstances governing the majority of adaptive responses, specific antigen binding heavily cross-links the B cell's surface immunoglobulin. This initial binding event triggers an intracellular cascade that

structurally primes the B cell and directs it to actively solicit requisite co-stimulation from an antigen-specific TH cell.

<FIGURE>

- (1) Extracellular antigen physically cross-links multiple BCR complexes. This physical clustering forces the spatial aggregation of the associated intracellular $Ig\alpha/Ig\beta$ signaling domains, leading to the rapid structural activation of their Immunoreceptor Tyrosine-based Activation Motifs (ITAMs).
- (2) Resident Src-family kinases bind these aggregated ITAMs, utilizing ATP hydrolysis to systematically phosphorylate critical tyrosine residues on the motifs.
- (3) The newly phosphorylated ITAMs serve as high-affinity docking sites for Syk kinase. The recruitment and activation of Syk subsequently initiates a vast, highly interconnected intracellular signaling web:
 - (a) DAG/IP3 Generation: Phospholipase C- γ (PLC- γ) is activated, cleaving the membrane phospholipid PIP2 into diacylglycerol (DAG) and inositol trisphosphate (IP3), acting as pivotal second messengers.

IP3 diffuses to trigger the rapid release of stored Ca^{2+} from the ER lumen into the cytosol. This calcium influx heavily binds calmodulin, which subsequently activates the phosphatase calcineurin to dephosphorylate and activate the transcription factor NFAT.

Concurrently, DAG activation propagates signals that result in the targeted degradation of the $I\kappa B$ inhibitory complex, thereby unleashing the potent pro-inflammatory transcription factor NF- κB to enter the nucleus.
 - (b) MAPK/ERK Kinase Cascade: Through intermediary adapter proteins, the classical RAS-RAF-MEK-ERK phosphorylation cascade is initiated. This culminates in the nuclear translocation of activated ERK proteins, which directly upregulate essential target genes controlling robust cellular proliferation and terminal differentiation.
 - (c) PI3K Activation: The activated BCR complex recruits PI3K, which phosphorylates residual membrane PIP2 to yield PIP3. The accumulation of PIP3 provides a powerful overarching survival signal, effectively blocking endogenous apoptotic pathways during the stress of rapid clonal expansion.
- (4) Driven by these integrated transcription factors, the fully activated B cells commence exponential mitotic division and begin differentiating into functional, antibody-secreting plasma cells.
- (5) Feedback Regulation: As the immune response progresses, the accumulating high-titer antigen-antibody immune complexes begin to bind inhibitory surface Fc γ receptors, specifically CD32 (Fc γ RIIB) which houses an ITIM domain, providing essential negative feedback to safely curtail the response. (Note: CD22 is an independent inhibitory coreceptor recognizing sialic acid, playing a similar dampening role).
 - (a) While strictly necessary for controlling inflammation, suppressing active signaling presents a risk of inducing peripheral tolerance toward foreign antigens—a state where the immune system inappropriately acknowledges a foreign entity without mounting a defense. This is heavily counterproductive during active pathogenic infections.
 - (b) To carefully manage this balance, specialized Regulatory T (Treg) cells act primarily to enforce systemic tolerance specifically directed against autologous host proteins, as well as maintaining non-reactivity toward benign, commensal microbiome flora.
 - (c) Clinically, if exogenous antibodies targeting a specific antigen are passively administered into a host system, they will effectively sequester the circulating antigen load, largely preventing it from natively binding host BCRs and thus actively suppressing the generation of a de novo immune response.

Core Concept 2.4: T_H Cell Synapse

Despite robust BCR engagement, sustained proliferation heavily relies upon physical contact with an activated TH cell. Notably, the high affinity of the BCR allows the B cell to capture and present antigen to TH cells at concentrations roughly 100 to 10,000 times lower than what is required for standard presentation by macrophages or dendritic cells.

- (1) BCR engagement triggers receptor-mediated endocytosis, allowing the B cell to internalize the bound antigen and subsequently hydrolyze it into small peptide fragments via the endosomal pathway.
- (2) These processed antigenic peptides are then loaded onto nascent MHC class II molecules and shuttled back to the external cell surface for presentation.
- (3) An activated TH cell specific for that exact peptide epitope binds the MHC II complex via its TCR, heavily stabilized by the CD4 co-receptor, forming a highly organized cellular interface termed the immunological synapse.
- (4) This direct recognition event induces the TH cell to rapidly upregulate surface expression of CD40 Ligand (CD40L/CD154), which securely engages the constitutively expressed CD40 receptor physically present on the B cell surface.
- (5) The CD40-CD40L interaction delivers the critical, definitive second signal to the B cell. This massively amplifies the internal signaling cascades initiated by the BCR, fully committing the B cell to exponential proliferation and terminal effector differentiation.

In this reciprocal exchange, the B cell concurrently upregulates surface B7 molecules (CD80/CD86), which engage CD28 on the TH cell to continually support and reinforce the TH cell's activated state.

2.2.2 Antibody Development**Core Concept 2.5: Lymph Node Follicle Development**

<FIGURE>

Circulating free antigen and small immune complexes present in the plasma or lymph are predominantly captured by local antigen-presenting cells (APCs).

These APCs traffic to the lymph node to initially activate TH cells within the paracortical region. These primed TH cells then migrate toward the B cell zones, forming functional immunological synapses with antigen-bearing B cells. This interaction generates a proliferating cellular cluster localized precisely at the boundary delineating the paracortex and the outer cortex.

This active cluster rapidly expands and reorganizes, driving the transformation of a resting primary follicle into a highly active secondary follicle, distinctly characterized by a well-defined light germinal center and a dense dark zone.

Germinal Center: The primary structural environment facilitating the intense interactions among B cells, specialized TH cells, and FDCs required for rigorous antibody selection.

Dark Zone: The anatomical location dedicated to extreme B cell proliferation accompanied by targeted somatic hypermutation.

NOTE: Within this environment, FDCs trap and stabilize intact antigen-antibody immune complexes within distinct beaded cellular structures termed iccosomes, presenting them efficiently for selection by competing B cell clones.

Core Concept 2.6: Affinity Maturation

<FIGURE>

Centroblasts—highly activated, morphologically enlarged B cells—migrate into the dark zone where they experience intense mitotic proliferation. Concurrently, they undergo somatic hypermutation, introducing targeted nucleotide substitutions strictly within the DNA sequences coding for the CDR loops. Throughout this proliferative stage, centroblasts temporarily cease the surface expression of their BCRs.

Upon exiting the cell cycle, these cells mature into centrocytes and migrate toward the light zone for a rigorous selection process. Centrocytes must actively compete to bind the limited antigen presented on FDC iccosomes. Cells possessing mutated receptors with improved affinity successfully bind the antigen, enabling them to present peptide to TH cells, thereby securing the vital CD40 survival signals. This process functions essentially as accelerated, localized natural selection.

Centrocytes bearing low-affinity or non-functional mutations fail to secure sufficient antigen or TH cell help; consequently, they undergo rapid apoptosis and are cleared locally by tingible body macrophages.

The highly selected, high-affinity centrocytes finally exit the germinal center microenvironment, terminally differentiating into long-lived plasma cells. These cells permanently downregulate surface receptor expression in favor of massive, sustained secretion of high-affinity antibodies targeted against the infection.

Core Concept 2.7: Class Switching

TH cells maintain synaptic contact with differentiating B cells, secreting tailored cytokine profiles that chemically direct class-switch recombination. This mechanism physically targets specific "switch regions"—conserved sequences spanning 2 to 3 kilobases positioned immediately upstream of each respective constant gene segment.

Prior to any switching event, the mature B cell exclusively transcribes the proximal μ and δ constant regions, forming its initial IgM/IgD receptors.

The switching mechanism involves the permanent excisional recombination of DNA, physically looping out and discarding the previously utilized constant domains alongside any intervening sequences positioned between the rearranged VDJ domain and the newly targeted downstream constant region.

Because this deletion occurs at the DNA level, any excised upstream constant regions are irretrievably lost from that allele's genome. Therefore, progressive class switching is unidirectional down the heavy chain locus.

This large-scale genetic remodeling generally requires a sustained immune response, typically occurring functionally prominent after a week or more of persistent antigen challenge.

Consequently, the primary serological hallmark of a novel acute infection is a rapid spike in low-affinity IgM; the transition to high-affinity IgG (or other specialized isotypes) is a slower, distinctly subsequent event characterizing the later phases of the adaptive response.

<FIGURE>

2.2.3 Plasma vs Memory B Cells**Core Concept 2.8: Plasma Cells**

The terminal differentiation from an activated centrocyte into a highly specialized plasma cell is orchestrated by key transcription factors. This shifts the cellular machinery, altering RNA splicing enzymes to selectively exclude the membrane-anchoring exons from the heavy chain transcript, thereby converting receptor expression into large-scale soluble secretion.

This terminal transition is strongly correlated with, and facilitated by, the significant downregulation and suppression of BSAP (Pax-5) expression.

Core Concept 2.9: Memory Cells

Conversely, cells diverted from terminal differentiation to become long-lived memory B cells possess a quiescent morphology superficially resembling naive B cells. However, they bear the genetic scars of class switching and affinity maturation, expressing specialized, high-affinity isotypes (such as IgG, IgA, or IgE) rather than the baseline IgM/IgD phenotype restricted to naive cohorts.

The stabilization and maintenance of the memory cell lineage is highly dependent on the continued, elevated expression of the BSAP transcription factor.

2.3 Leukemia & Lymphoma

Core Concept 2.10: Leukemia & Lymphoma

The Burden of Non-Productive Rearrangements

The highly error-prone and structurally randomized nature of the V(D)J recombination process introduces immense genomic instability.

Statistically, the imprecise resolution of coding joints possesses roughly a 66% probability of introducing a catastrophic frameshift mutation during any single recombination event.

Furthermore, the stochastic action of Artemis (P-nucleotides) and TdT (N-nucleotides) frequently generates premature, non-sense stop codons directly within the reading frame.

Failing to generate structurally viable heavy and light chains results in a non-productive rearrangement, inevitably condemning the developing progenitor cell to immediate apoptosis.

Allelic Exhaustion

A developing cell will exhaust all available allelic options—attempting rearrangement on both heavy chain alleles and across the four total light chain alleles—before finally succumbing to apoptotic clearance.

This intense requirement for repeated, double-stranded DNA breaks and rampant cellular proliferation inherently predisposes the B cell lineage to oncogenic transformation. Erroneous recombination events, such as translocations involving the highly active Ig promoters and critical proto-oncogenes, form the fundamental molecular basis for many lymphomas and leukemias.

<FIGURE>

3 Complement System

3.1 MAC Complex Attack

Core Concept 3.1: MAC Complex Attack

<FIGURE>

This process is called membrane lysis (poking holes in pathogenic membranes)

The MAC pathway can be triggered through both innate and adaptive recognition—this is the last stage of the complement activation pathway (all complement pathways converge to the cleavage of C5)

The MAC complex is essentially a ring of C9 peptides that serves as a passage hole for harmful substances to degrade the insides of cells like...

membrane coated viruses

rough gram negative (membrane-peptidoglycan-membrane); gram positive is difficult because of its thick walls

eukaryotic parasites

erythrocytes or RBCs (under pathogenic conditions)

cancer cells

The size of the MAC complex is dependent on the number of C9s that forms the ring (70 to 100)

Most complements are synthesized by the liver and release in an inactive form (VERY SIMPLIFIED)

3.2 Complement Pathways

Core Concept 3.2: Classical Pathway

<FIGURE>

An inactive C1 complex—composed of 6 C1q (C1qA, C1qB, C1qC), 2 C1r, and 2C1s subunits—activates when at least 2 of the C1q subunits' heads bind to antibodies, making the C1rs complex wrap on the C1q stems and exposing the C1rs complex's catalytic activity'

The oligosaccharide and the Ig domain it is attached to constitute the principal agent of activation

Activating Antibody Isotypes

IgM—very effective—usually floats in a planar conformation within plasma, but when binded to an antigen, it protrudes its Fc stem outwards (making it easier to bind)

IgG is monomeric. The C1 complex needs two IgGs to bind to in order to go down its pathway (this prevents random activation)

IgG3 is the best, IgG1 is the second best, IgG2 barely activates, IgG4 doesn't work at all

IgA, IgE, and IgD do not activate the complement system

<FIGURE>

Core Concept 3.3: Lectin Pathway

The Lectin pathway looks the same as the Classical pathway except these few points

<TABLE>

The rest of the pathway starting from the cleavage of C4 and C2 are the same

Core Concept 3.4: Alternative Pathway

<FIGURE>

Core Concept 3.5: Side Signal Pathways & Other

Intro sentence

(a) Inflammation (*Anaphylatoxin*)

- *C3a*: Triggers degranulation of mast cells and basophils, with also permeability changes in the endothelium
- *C5a*: Summons macrophages and neutrophils

(b) Coating

- *Opsonization*: Complements bind to pathogenic cells, enhancing phagocytosis towards them—[C3b]
- *Viral Neutralization*: Simply attaching a complement to a host cell invader makes it much more difficult to invade
- *Immune Complex Clearing*: Immune complexes (ICs) like antigen-antibody and antigen-complement complexes tend to bind to RBCs and get filtered as then circulate

4 Major Histocompatibility Complex

4.1 MHC Structure & Genetics

Core Concept 4.1: MHC Characteristics

<FIGURE>

<TABLE>

Core Concept 4.2: MHC Gene Sequences

<FIGURE>

The MHC gene sequences is found in chromosome 6

This whole block of genes is termed a haplotype

MHC in humans is commonly referred to as Human Leukocyte Antigen (HLA) and is part of the immunoglobulin superfamily

REMINDER: there are 2 copies of these gene sequences (one from mom, one from dad)

(a) MHC I

Further separated by three different versions: HLA-A, HLA-B, HLA-C (each with a different specificity for peptide binding)

There are currently 68 HLA-A, 125 HLA-B, and 44 HLA-C version within the human population that have been identified (thus an underestimate)

An individual will express at max 6 different types of MHC I proteins—assuming that the maternal alleles are distinct from the paternal alleles

***This gene sequence only covers the α subunits; the β subunit locus is on chromosome 15 (non-polymorphic)

WILDLY POLYMORPHIC (high allele counts for HLA versions)

NO ALTERNATIVE SPLICING FORMS (DNA is not changed)

(b) MHC III

This gene sequence is completely different from the main MHC proteins I and II

It is included just because it happens to lie in between

Codes for secreted proteins involved in innate response: [complement proteins], [cytokines], [heat shock proteins]

(c) MHC II

Each subsection—HLA-DP, HLA-DQ, HLA-DR (each with a different specificity for peptide binding)—codes for 2 separate peptides (α and β) which function together to make up the MHC II protein

An individual will express at max 12 different types of MHC II proteins (due to potential paternal-maternal α - β dimers)—assuming that the maternal alleles are distinct from the paternal alleles

WILDLY POLYMORPHIC (high allele counts for HLA versions)

NO ALTERNATIVE SPLICING FORMS (DNA is not changed)

(d) Other

Interspersed in the MHC II genes are those that code for elements of the antigen processing system for MHC I

There are other MHC I proteins involved in other functions (specialized receptor, transport, presentation)

Core Concept 4.3: MHC Mendelian Genetics**(a) Non-Recombinant Inheritance**

<FIGURE>

The alleles coding for the MHC gene sequences tend to be passed on to offspring as a block (with no crossing over; tightly linked)—thus, in most cases, siblings will have a $\frac{1}{4}$ chance to have the same MHC genes

**The term “haplotype” is used in other contexts where a group of genes remains in a block and be passed on as a unit (e.g. & chromosomes and Mitochondrial DNA)

(b) Recombinant Inheritance

<FIGURE>

Rarely, if the cross over happens between the I and II block, this produces a recombinant haplotype (highly unlikely to have sibling match)

This increases variability into the gene pool

Inbreeding

With inbreeding—there will be no introduction of new haplotypes of the MHC gene sequences within the population—thus each individual will have the same haplotype on each homologous chromosome

This individual will thus have one 3 versions of MHC I and 3 versions of MHC II rather than the maximum 6 and 12 respectively

4.2 MHC Antigen Preparation

Core Concept 4.4: Endogenous MHC I (T_c Cells)

(1) Proteasomal Degradation of Antigen to Peptide

<FIGURE>

The antigen is targeted for degradation by binding to ubiquitin—a small protein—and is sent to the interior of a proteasome

Proteasome is made up of a protein with a 7-fold symmetry (associated with danger) and is more active under conditions of stress and damage by their nature

LMP2 and LMP7 replace proteasomal subunits under inflammatory conditions and promote degradation into the correct peptide size (9 amino acids)

There are also complex backup systems for this

LMP2 and LMP7 genes are located within the MHC II gene sequence and are polymorphic

(2) Transport to the RER

<FIGURE>

The TAP1 and TAP2 bind to the peptides, hydrolyze ATP, and drive them into the RER lumen

TAP1 and TAP2 genes are located within the MHC II gene sequence and are polymorphic

TAP Protein Structure

Heterodimer

Cytosolic binding side for peptides

ATP hydrolysis site

Hydrophobic anchoring regions with 3 switchbacks per peptide

(3) Assembly of MHC I

(a) Calnexin- α to α - β

<FIGURE>

The α chain is inserted into the membrane, and the β microglobulin chain sent through to the lumen

Here calnexin ensures the α chain's proper folding

Once the β microglobulin chain bind to the α chain, calnexin unbinds

(b) α - β -Calreticulin-Tapasin-TAP-ERp57

<FIGURE>

The complex associates within calreticulin and tapasin, and then ERp57 binds over the cleft

Calreticulin and ERp57 both help stabilize the molecule and prepare it for peptide binding

Tapasin brings over the TAP complex over to the pre-peptide MHC I

(c) α - β -Calreticulin-Tapasin-TAP-Peptide

<FIGURE>

TAP protein hands off the peptide via tapasin to the cleft which displaces ERp57

(d) α - β -Peptide to Golgi

<FIGURE>

Calreticulin and tapasin dissociate from the MHC I-peptide complex and travel to the Golgi

(4) Transportation to Exterior

<FIGURE>

After final sorting the Golgi, the vesicle with MHC I-peptide complex is released by exocytosis onto the membrane exterior

Core Concept 4.5: Exogenous MHC I / Cross Presentation (T_c Cells)

(1) Antigen Uptake

<FIGURE>

Through endocytosis, the cell picks up the exogenous antigen

(2) Proteasomal Degradation of Antigen to Peptide

<FIGURE>

After getting degraded within the endosome through degradation stages, the peptide fragments travel to and fuse with the RER, exposing the exogenous peptides to MHC I proteins

NOTE: This process only happens in APCs, and activated TC cells do not attack the APCs because they don't express co-stimulatory danger signals that promote an attack on the presenting cell. The purpose of cross presentation is to lower the response time of TC cells.

Core Concept 4.6: Exogenous MHC II

(1) Antigen Processing

(a) Antigen Uptake

<FIGURE>

The antigen is internalized through phagocytosis—a specialized form of endocytosis during which there occurs an engulfment of whole particles with pathogen

The formed vesicle in the cell interior is named an endosome

(b) Hydrolysis in Vesicles

<FIGURE>

An endosome goes through three stages (each one more acidic and hydrolytic) and use lysosomes to degrade the foreign antigens (and remove carbohydrates and lipids), cutting them up to their proper size (13-18 amino acids long)

Early endosome, pH 6.0 - 6.5

Late endosomes or endolysosomes, pH 5.0 - 6.0

Lysosomes (or phagolysosomes), pH 4.5 - 5.0

(2) MHC II

(a) Preparation of MHC II

<FIGURE>

After translation of the MHC II $\alpha\beta$ heterodimer onto the RER membrane, the heterodimer associates with the invariant chain (Ii)

The figure only shows one invariant chain binding, but in reality, the Ii comes as a trimer—three MHC II heterodimers bind at a time

The invariant chain [1] blocks the cleft (to prevent immature binding), [2] ensures proper conformation, [3] helps RER exiting, and [4] signals for fusing with phagolysosomes

(b) *Transportation of Golgi*

<FIGURE>

The $\alpha\beta$ -Ii complex is then transported to the Golgi where it awaits fusion with a phagolysosome

(3) **Fusion with Endocytic Vesicles/Phagolysosomes**

<FIGURE>

The $\alpha\beta$ -Ii complex fuses with early endosomes after leaving the Golgi

(4) **Invariant Chain Degradation**

<FIGURE>

After the development of the endosomes into more acidic internal environments, Ii gets degraded, leaving only the CLIP attached to the MHC II and still blocking the cleft region

(5) **CLIP Removal**

<FIGURE>

At the end of antigen degradation into peptides, non-classical MHC (HLA-DM) removes the CLIP, allowing for peptide binding and the formation of a peptide-presenting MHC II protein (which is very stable)

(6) **Transportation to Exterior**

<FIGURE>

The peptide-MHC II complex is then transported to the cell surface to be displayed

Core Concept 4.7: Exogenous CD1 MHC(1) **Glycolipid Uptake**(2) **Preparation and Expression of CD1 MHC**(3) **Retreat of CD1 MHC**(4) **Binding of CD1 MHC to Glycolipid**(5) **Transportation to Exterior**

NOTE: Several different types of cells and receptors can bind to and respond to the CD1 lipid presentation

T cells bind to the lipid-CD1 MHC complex through the $\gamma\delta$ receptor (with no CD4 or CD8)

TH cells can also bind through the $\alpha\beta$ receptors using CD4

NK cells

invariant natural killer T cells (iNKT) with CD1

5 T Cells

5.1 T Cell Receptor

5.1.1 TCR Complex Characteristics

<FIGURE>

Core Concept 5.1: $\alpha\beta$ and $\gamma\delta$ Receptors

<FIGURE>

T cells can make either the $\alpha\beta$ or $\gamma\delta$ TCR—a heterodimer (most T cells make the $\alpha\beta$ version)

Both resemble a truncated Ig and is part of the Ig superfamily with three hypervariable loops in each chain (CDRs)

The positively charged transmembrane regions of the TCR and negatively charged transmembrane regions of the CD3 signal transducers form a salt bridge attraction that holds the complex together

During development, the TCR does not go through affinity maturation like B cells, thus they need the supplement receptors (CD4/8) in order to strengthen this less specific binding

$\alpha\beta$ Receptor

Undergoes selection to produce a highly specific recognition molecule

$\gamma\delta$ Receptor

Undergoes selection to produce a quasi innate recognition molecule (some specificity but not as much as $\alpha\beta$)

Mostly for recognizing lipid antigens of bacteria and parasites

Important in patrolling mucosa and epithelium

Core Concept 5.2: $\gamma\epsilon$ and $\delta\epsilon$ Heterodimers

<FIGURE>

All chains (γ , δ , ϵ) are part of the Ig superfamily

All peptides have an immunoreceptor tyrosine-based activation motif (ITAM) at the cell-interior side of the receptor

Core Concept 5.3: $\zeta\zeta$ or $\zeta\eta$ Heterodimers

<FIGURE>

Chains are NOT part of the Ig superfamily

Only 9 amino acids to the exterior and a long extended cytoplasmic tail

Has 3 ITAMs rather than one

5.1.2 CD4 & CD8**Core Concept 5.4: CD4**

<FIGURE>

Used by TH cells to bind to MHC II

Monomer

4 extracellular Ig domains

N-terminal binds to the base of MHC II (junction of $\alpha 2$ and $\beta 2$ domains) at a hydrophobic pocket

C-terminal possess 3 serine regions that can be phosphorylated

Core Concept 5.5: CD8

<FIGURE>

Used by TC cells to bind to MHC I

Dimer— α - β hetero or α - α homo linked by disulfide

1 extracellular Ig domain per peptide

N-terminal binds to the $\alpha 2$ and β microglobulin

C-terminal possess amino acid regions that can be phosphorylated

5.1.3 TCR Genes and Rearrangement

Core Concept 5.6: Gene Sequence and Receptor Overview

<FIGURE>

(a) α & δ Peptide

<FIGURE>

Chromosome 14 Long Arm

Alpha (α): Resembles the light chain Ig gene because of absence of D regions $40 V - N/A D - 50 J - 1 C$

Delta (δ): Resembles the heavy chain Ig gene because of presence of D regions $3 \text{ (or more)} V - 3 D - 4 J - 2 C$

(b) β Peptide

<FIGURE>

Chromosome 7 Long Arm

Resembles the heavy chain Ig gene because of presence of D regions $40 V - 2 \text{ (functional)} D - 6 \text{ and } 7 J - 2 C$

(c) γ Peptide

<FIGURE>

Chromosome 7 Short Arm

Resembles the light chain Ig gene because of absence of D regions $5 \text{ (functional)} V - N/A D - 5 J - 2 C$

Core Concept 5.7: α Rearrangement

(1) Gene Sequence

<FIGURE>

(2) δ Removal & VL-J Joining

<FIGURE>

**Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

(3) Result

<FIGURE>

[Variable # of VL] - [VL] - [J] - [Variable # of J] - [Intron] - [C]

Next is EXPRESSION

Core Concept 5.8: β Rearrangement**(1) Gene Sequence**

<FIGURE>

(2) VL-D/J Joining

• VL-DJ

<FIGURE>

VL joins with a D

**Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

• VL-J

<FIGURE>

VL joins with a J (all Ds get cut out)

**Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

(3) Result

• VL-DJ

<FIGURE>

• VL-J

<FIGURE>

Next occurs EXPRESSION

Core Concept 5.9: γ Rearrangement**(1) Gene Sequence**

<FIGURE>

(2) VL-D/J Joining

<FIGURE>

**Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

(3) Result

<FIGURE>

Next occurs EXPRESSION

Core Concept 5.10: δ Rearrangement**(1) Gene Sequence**

<FIGURE>

(2) VL-D/J Joining & Result

<FIGURE>

The VL sequence either joins to a D or J depending on what it splices out

NOTE: This rearrangement can add more than one D to the sequence as well

****Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability**

Next occurs EXPRESSION

Core Concept 5.11: Subsequent Expression**(1) The message gets transcribed**

NOTE: TCR C regions always has its bulk C gene sequence followed by an extracellular gene (XC), a transmembrane gene (TM), and a cytoplasmic tail gene (CA)

<FIGURE>

(2) The message precursor is capped and gets a poly-A-tail**(3) Processing occurs (removing the introns and other extra Js)**

<FIGURE>

(4) Translation of the L region attaches the ribosome to the RER and inserts into the lumen with the C-terminal bound to the ER membrane**(5) The L region is clipped off****(6) Enzymes in the Golgi form disulfide bonds****(7) Vesicles with finished TCRs get transported to and fuses with the plasma membrane****Core Concept 5.12: History: Hendrick and Davis**

Hendrick and Davis were responsible for the gene identification of TCRs

(1) Selecting Membrane Bound Polysomes

<FIGURE>

H&D first separated membrane bound from cytoplasmic proteins by isolating mRNA from membrane bound polysomes

T cells—TCR, CD3, CD4/8

B cells—Ig receptor (BjkkCR), co-receptors

Both of these cells with have other things in common as well

(2) Subtractive Hybridization

<FIGURE>

H&D used membrane bound B cell genes to selectively hybridize with and remove any overlapping radioactive cDNA from T Cells, leaving only the radioactive cDNA unique to T cells behind

(3) Gene Comparison Across Multiple T Cells

<FIGURE>

Now there are only 10 different cDNA clones remained (unique to T cells) and we have to find which ones are rearranged (which would be the TCR)

We run each cDNA portion across multiple T cells lineages

If the cDNA portion (for example, g1) is not rearranged, the length traveled in the gel electrophoresis would be the same across all T cell lineages and B cells (control) as well

If the cDNA portion (for example, g2) is rearranged, the length traveled in the gel electrophoresis would be different across different T cell lineages because of their variable lengths

The rearranged gene were 2 out of the 10 cDNAs, thus locating the genes for the TCR

Alloreactivity is T cells being able to bind with foreign MHCs, which it should not be able to do (Figure on MHC Characteristics)

5.2 T Cell Development

Core Concept 5.13: Developmental Pathway

<TABLE>

5.3 TCR Antigen Response

Core Concept 5.14: Antigen Response Pathway

(1) Antigen Recognition

APC presents an antigenic peptide on MHC II

(2) Binding

TH cell binds to this complex through TCR-CD3 complex, aided by CD4

*Superantigens are viral or bacterial proteins that nail together TCR-antigen-MHC complexes, hyperstimulating the TH cells and overproducing cytokines (which could kill you)

(3) Signaling Cascade

A signaling cascade starts towards the nucleus and pushes the cell into cell division

(4) Transcriptional Activation

- *Immediate Genes*

Genes for transcription factors for inflammatory response

—NFAT, AP-1, NF-κB

- *Early Genes*

Cytokines, Interleukins and their receptors

*This is where developmental branching occurs, and the TH cells differentiate into either TH1, TH2, THreg, or TH17 (sometimes referred to as TH3)

Inducing cytokines are produced by other immune cells like macrophages and DCs which during the

production of master transcription factors

—[Type]: [inducing cytokine] → [master transcription factor]

—Th1: IL-12 → T-bet

—Th2: IL-4 → GATA3

—Th17: TGF- β + IL-6 → ROR γ t

—Treg: TGF- β + IL-2 → FOXP3

- *Late Genes*

Cell adhesion molecules (CAMs) and effector cytokines

Core Concept 5.15: TCR Antigen Recognition to ZAP70

<FIGURE>

(1) APCs present their MHC II and antigen complex to the TCR

(2) CD4 helps strengthen this binding

(3) p56lck (which is associated with CD4/8 on its cytoplasmic tail) phosphorylates the ITAMs, providing a docking site for ZAP 70

(4) ZAP 70 docks on the ITAMs and get phosphorylated by p56lck and activated

(5) ZAP-70 goes on to phosphorylate a variety of molecules, setting off other downstream pathways through transcriptional activation

—DAG-IP3 formed through hydrolysis of membrane phospholipid by phospholipase C, causes ER to release Ca²⁺

—Ca²⁺ activates NFAT (up regulates immune response)

—DAG activates NF- κ B (up regulates immune response)

—Activation of Ras, the G protein

—Transcription factors activated by MAP kinases turn on genes that trigger cell division

Regulation

B7 is a membrane bound protein on APCs binding either to the CD28 or CTLA-4 of a T cell

CD28 Binding (Upregulation)

The cell will begin proliferating and dividing, transcribing the message for both IL-2 and the high affinity version of its TCR α chain and enter into a positive feedback loop.

*An APC without the B7 for CD28 would turn the T cell off

CTLA-4 Binding (Downregulation)

The T cell will begin to downregulate its development

*Resting T cells initially have no CTLA-4, but begin to produce it during activation

Core Concept 5.16: Lipid Rafts

Over the course of the signaling lipid rafts migrate towards the outside of the synapse region as a result of CAMs associating CD45 to CD22

- (1) TCR, lipid raft after antigen binding
<FIGURE>
- (2) Synapse during mid-activation
<FIGURE>
- (3) Synapse, mature (bull's eye formation)
<FIGURE>

6 Cytokines and Signals

6.1 Receptor Signaling Types

Core Concept 6.1: Immunoglobulin

<FIGURE>

Structure Receptors have β pleated sheet (Ig) domains and membrane span characteristics of family

Ligands—antigens, integrins, growth factors

Core Concept 6.2: Type I Receptors

<FIGURE>

Structure

Heterodimers or trimers with interchangeable β and γ subunits across different receptors

Composed of fibronectin domain (like Ig domains), two β pleated sheets running opposite

β, γ N terminals—Four conserved cysteines that stabilize the structure using 2 disulfide bonds

β, γ proximal domains—conserved [tryptophan]-[serine]-[whatever]-[tryptophan]-[serine] sequence (function unknown)

β, γ C terminals—domains have sites that recruit cytosolic tyrosine kinases (phosphorylation)

Ligands—a small monomer with 4 α helical domains

JAK-STAT Pathway—

Ligand binding brings JAKs together and activates

P-JAKs phosphorylate the cytoplasmic receptor tails (docking site for STAT)

JAKs phosphorylate the STATs, which migrate to the nucleus, and up-regulate other genes

Core Concept 6.3: Type II Receptors

<FIGURE>

Structure

β, γ heterodimers with extensive cytoplasmic signaling domains

Fibronectin domain

β, γ N terminals—CCCC that stabilize the structure using 2 disulfide bonds

Ligands—series of α helices

JAK-STAT Pathway—SAME as Type I

Ligand binding brings JAKs together and activates

P-JAKs phosphorylate the cytoplasmic receptor tails (docking site for STAT)

JAKs phosphorylate the STATs, which migrate to the nucleus, and up-regulate other genes

***JAKs and STATs have variety (4 and 6 known respectively) and the signaling process uses more than one version of each

Core Concept 6.4: Interleukin 17

<FIGURE>

Structure

Heterodimers, Homodimers, Trimers

IL17R composed of 2 IL17RAs and 1 IL17RC

Ligands—IL-17A: β pleated sheets held together by 2 S=S bonds constructed by CCCC (pro-inflammatory)

Activates the Act1 Pathway (NF- κ B, MAPK, CCAAT)

**IL-17 is secreted by Th17, iNKT, and $\gamma\delta$ T cells (mainly for border patrol)

Core Concept 6.5: Tumor Necrosis Factor

<FIGURE>

Structure

Four repeated β pleated sheet domains stabilized by CCCC's S=S bonds

Upon binding TNF, subunits cluster in trimer, leading to activation

Ligands—juxtacrine factors composed of β pleated sheets and displayed in clusters of three subunits

Pathway binds to Death Domain (DD), TRADD, FADD, and RIP → canonical NF- κ B pathway

Core Concept 6.6: Chemokines

<FIGURE>

Structure

7-spanning α helices in membrane associated with G-protein on C-terminal

Very promiscuous receptor (cross reception)

Ligands— α helices and β pleated sheets stabilized with CCCCs

G Protein Pathway—

Ligand binding to receptor induces α subunit of G protein to switch out its GDP to GTP

α , β , γ G Protein subunits all dissociate and start their individual signaling cascades

6.2 Immune Signaling Types & Themes

Core Concept 6.7: Immune Signaling Types

<TABLE>

Core Concept 6.8: Signaling Themes

<FIGURE>

(a) Phosphorylation

Adding phosphates to, and removing phosphates from molecules

Examples: BCRs, JAK-STAT, TCRs, G Proteins, Membrane inositols

(b) Phospholipid Hydrolysis

Phospholipids are split

Examples:

Phospholipid is split into

DAG (non polar part)

IP3 (polar part) cascades

*IP3 makes the ER leaky to Ca^{2+} , leading to the NFAT pathway through binding to calmodulin

(c) Protein Hydrolysis

Using hydrolysis of cytosolic components in situations where you don't want to reverse the signal (once you've clipped a protein, you can't reassemble it, you must synthesize a new one)

Examples: Notch Signaling, Apoptosis Induction

(d) Cytoskeletal Changes

In relation to the movement of a cell throughout the body

Examples: Chemokine $\rightarrow$ 7-spanR $\rightarrow$ G Protein signal $\rightarrow$ Integrin conformational change $\rightarrow$ Talins $\rightarrow$ Actin microfilaments $\rightarrow$ Movement

6.3 Cytokine Interaction Types

Core Concept 6.9: Cytokine Interaction Types

(a) Pleiotropy

<FIGURE>

One cytokine may act on several different cell types

IL-4 coordinates a Th2 response by acting on both B and mast cells

(b) Redundancy

<FIGURE>

More than one cytokine will produce the same effect

IL-2, 4, and 5 induces proliferation in B cells

(c) **Synergy**

<FIGURE>

Combinations of stimulus have effects neither has by itself

IL-4 and IL-5 together will cause B cells to class switch and make IgE immunoglobulins

(d) **Antagonism**

<FIGURE>

One cytokine block the effects of another

IFN γ (a Th1 promoter) blocks the ability of IL-4 (a Th2 promoter) to induce class switching to IgE

(e) **Cascade Inductionk**

<FIGURE>

One cytokine stimulates other cells to produce a different cytokine, which stimulates more cells

Macrophage $\rightarrow$ IL-12 $\rightarrow$ Th1 $\rightarrow$ IFN γ $\rightarrow$ activated macrophage pathway of development